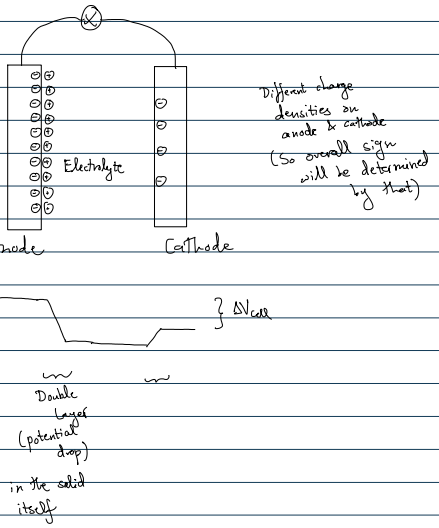
Lecture 20 (CHE)

Tuesday, 18 April 2023 10:32 AM

Electrochemistry

- electrocatalyst
- batteries
- capacitors
- devices
- Corrosion

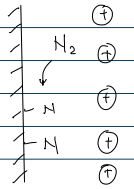


Key features: ① Solvent/solid interfaces
water - electrocatalysis
organic - batteries

② Existence of a charged double layer near the surface of an electrode that establishes potential drops across surfaces

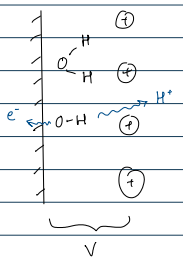
Reactions @ electrified interfaces

Chemical:



+ No e^- transfer
+ Small voltage dependence (rearrangement of anions)
↓
Can use normal DFT (possibly including solvent molecules)

Electrochemical:



$* + H_2O \rightarrow OH^* + H^+ + e^-$

→ Generate positive/negative charges and so are hugely affected by voltage

Analyzing thermodynamics.

+ Calculating ΔG:

$$\Delta G = \mu_{OH^*} + \mu_{H^+} + \mu_{e^-} - \mu_* - \mu_{H_2O}$$

$$\mu_{OH^*} \approx E_{OH^*} - TS_{OH^*} + ZPE_{OH^*} + \dots$$

$$\mu_* \approx E_*$$

$$\mu_{H_2O} \approx E_{H_2O} - TS_{H_2O} + ZPE_{H_2O}$$

$\mu_{H^+} + \mu_{e^-}$: assume proton entropy (activity) is $\sim k_B T \ln[H^+]$ } Assuming that H^+ don't interact w/ each other → even at pH 0, 1 H^+ per 55 H_2O molecules [So pretty good assumption unless negative pH's]

+ Assume electrostatic energy to separate charges across double layer is $\sim qV = -eV$

Thus, $\mu_{H^+} + \mu_{e^-} = \text{const.} + k_B T \ln[H^+] - eV$

(const.: undetermined)

Thus, $\mu_{H^+} + \mu_{e^-} = \text{const.} + k_B T \ln[H^+] - eV$

(const. ← undetermined)

Standard conditions: $\mu^\circ_{H^+} + \mu^\circ_{e^-} = \text{const.} + k_B T \ln[H^+]_0 - eV$

$$\therefore \boxed{\mu_{H^+} + \mu_{e^-} = \mu^\circ_{H^+} + \mu^\circ_{e^-} + k_B T \ln \frac{[H^+]}{[H^+]_0} - e(V - V_0)}$$

V_0 is an arbitrary reference (std state) for voltage; often chosen to be: $-eV_0 \equiv \frac{1}{2}\mu^\circ_{H_2(g)} - \left(\mu^\circ_{H^+} + \mu^\circ_{e^-}\right)$ } Standard hydrogen electrode reference

↳ Experimentally, use a counter electrode where $[H_2^\circ]$ and $[H^+]_0$ are in equilibrium w/ electrode surface

$H^\circ_{2(g)}$: typically 1 bar

$H^{+\circ}$: typically 1M

We get, $$\boxed{\mu_{H^+} + \mu_{e^-} = \frac{1}{2}\mu^\circ_{H_2} + k_B T \ln \frac{[H^+]}{[H^+]_0} - eV_{SHE}} \quad ; \quad \mu^\circ_{H_2} = E_{H_2(g)} - TS^\circ_{H_2(g)} + ZPE_{H_2(g)}$$

($[H^+]_0$ ← related to pH)

V_{SHE} has the μ_{H^+} dependence

$\Delta G_2 = \Delta G_2(V, pH, BE)$

Example: ORR

1) $O_{2(g)} + 2* \leftrightarrow 2O*$

2) $O* + H^+ + e^- \leftrightarrow OH*$

3) $OH* + H^+ + e^- \leftrightarrow * + H_2O_{(l)}$

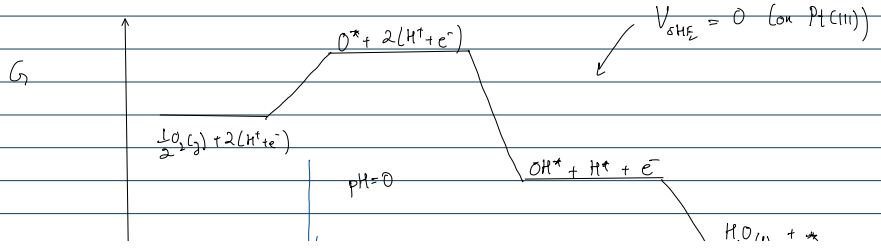
$\Delta G_1 = 2\mu_{O*} - 2\mu_* - \mu_{O_{2(g)}}$ [Independent of V]

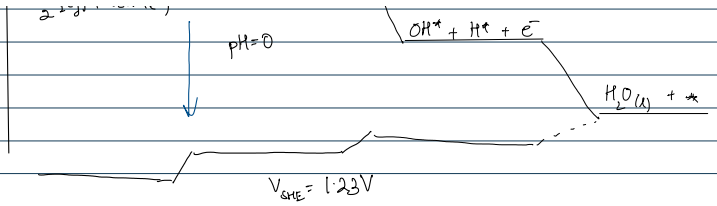
$\Delta G_2(V, pH) = \mu_{OH*} - \mu_{O*} - \left(\frac{1}{2}\mu^\circ_{H_2} + k_B T \ln \frac{[H^+]}{[H^+]_0}\right.$

$= \Delta G_2(0,0) - k_B T \ln \frac{[H^+]}{[H^+]_0} + eV_{SHE}$

$\Delta G_3(V, pH) = \mu_{H_2O(l)} + \mu_* - \mu_{OH*} - \left(\frac{1}{2}\mu^\circ_{H_2} + k_B T \ln \frac{[H^+]}{[H^+]_0} - eV_{SHE}\right)$

All quantities can be estimated w/ DFT → make a free energy diagram





V_{SHE} 0V : O_2 dissociation is RLS

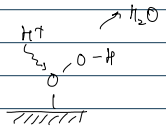
V_{SHE} 1.23V : conversion of OH^* to water may be RLS

} RLS can change w/ voltage in electrochemical systems. [RLS may be voltage dependent]

The potential at which all electrochemical steps are thermodynamically energetic is called the onset potential } approximation of V where rate becomes non-negligible

↳ We know this because when you lower the voltage from 1.23V there will be a point where the plateaus are level. (doesn't mean that they are at equilibrium)

Alternate ORR mechanism:



1) $O_{2(g)} + H^+ + e^- + * \longleftrightarrow OOH^*$

2) $OOH^* + H^+ + e^- \longleftrightarrow O^* + H_2O_{(l)}$

3) $O^* + H^+ + e^- \longleftrightarrow OH^*$

4) $OH^* + H^+ + e^- \longleftrightarrow H_2O_{(l)} + *$

Overall: $O_{2(g)} + 4(H^+ + e^-) \longleftrightarrow 2H_2O$

$$\Delta G_1(V_{SHE}, pH) = \mu_{OOH^*} - \mu_* - \mu_{O_2(g)} - \left(\frac{1}{2}\mu^\circ_{H_2} + k_B T \ln\frac{H^+}{H^+_\circ} - eV_{SHE}\right)$$

$$= \Delta G_1(0,0) + eV_{SHE} \quad @\ pH=0$$

$$\Delta G_2 = \mu_{O^*} + \mu_{H_2O(l)} - \mu_{OOH^*} - \left(\frac{1}{2}\mu^\circ_{H_2} - eV_{SHE}\right)$$

$$\Delta G_3 = \mu_{OH^*} - \mu_{O^*} - \left(\frac{1}{2}\mu^\circ_{H_2} - eV_{SHE}\right)$$

$$\Delta G_4 = \mu_{H_2O(l)} + \mu_* - \mu_{OH^*} - \left(\frac{1}{2}\mu^\circ_{H_2} - eV_{SHE}\right)$$

Ex: $\Delta G^\circ_2(0,0) = \mu^\circ_{O^*} + \mu^\circ_{H_2O(l)} - \mu^\circ_{OOH^*} - \frac{1}{2}\mu^\circ_{H_2}$

$\mu^\circ_{H_2O(l)} = \mu^\circ_{H_2O(g)} - \Delta G^\circ_{vap}$ ← from exp.

$= E_{H_2O(g)} - TS^\circ_{H_2O(g)} + ZPE_{H_2O} - \Delta G^\circ_{vap}$

; $\mu^\circ_{O^*} = E_{O^*} + ZPE_{O^*} - TS^\circ_{O^*}$

$\mu^\circ_{OOH} = E_{OOH^*} + ZPE_{OOH^*} - TS^\circ_{OOH^*}$

↳ from exp.

$= E_{H_2O(g)} - TS^\circ_{H_2O(g)} + ZPE_{H_2O(g)} - \Delta G^\circ_{vap}$

$\mu^\circ_{OOH} = E_{OOH^*} + ZPE_{OOH^*} - TS^\circ_{OOH^*}$

$\mu^\circ_{H_2} = E_{H_2(g)} + ZPE_{H_2(g)} - TS^\circ_{H_2(g)}$

$$\Delta G_2^\circ = \left(E_{O^*} + E_{H_2O(g)} - E_{OOH^*} - \frac{1}{2}E_{H_2(g)}\right) + \left(ZPE_{O^*} + ZPE_{H_2(g)} - ZPE_{OOH^*} - \frac{1}{2}ZPE_{H_2(g)}\right)$$
$$- T\left(S^\circ_{O^*} + S^\circ_{H_2O(g)} - S^\circ_{OOH^*} - \frac{1}{2}S^\circ_{H_2(g)}\right) - \Delta G^\circ_{vap}$$
$$= \Delta E^\circ_{DFT} + \Delta ZPE - T\Delta S^\circ - \Delta G^\circ_{vap}$$

Often, define O^* and OOH^* w/ respect to a common reference: $\Delta E_O \equiv E_{O^*} + E_{H_2(g)} - E_{H_2O(g)} - E_*$

$[H_2O(g) + * \leftrightarrow O^* + H_2(g)]$

$\Delta E_{OOH} = E_{OOH^*} + \frac{3}{2}E_{H_2(g)} - 2E_{H_2O(g)} - E_*$

$\left[2H_2O(g) + * \leftrightarrow OOH^* + \frac{3}{2}H_2(g)\right]$

$$\boxed{\Delta G_2^\circ = \Delta E_O - \Delta E_{OOH} + \Delta ZPE_2 - T\Delta S_2^\circ - \Delta G^\circ_{vap}}$$

PCCP: 2007

$\Delta E_{OOH} = 0.53\,\Delta E_O + 3.18\ eV$

$\Delta E_{OH} = 0.5\,\Delta E_O + 0.05\ eV$ } RPBE

$\Delta G_2^\circ = \underbrace{0.47\,\Delta E_O}_{\text{depends on catalyst}} \underbrace{- 3.18\ eV + \Delta ZPE_2 - T\Delta S_2^\circ - \Delta G^\circ_{vap}}_{\text{ind. of catalyst}}$

$\Delta G_1^\circ = 0.53\,\Delta E_O + 3.18\ eV + \left(2E_{H_2O(g)} - E_{O_2(g)} - 2E_{H_2(g)}\right) + \Delta ZPE_1 - T\Delta S_1^\circ$

$\Delta G_3^\circ = -0.5\,\Delta E_O + 0.05\ eV + \Delta ZPE_3 - T\Delta S_3^\circ$

$\Delta G_4^\circ = -0.5\,\Delta E_O - 0.05\ eV + \Delta ZPE_4 - T\Delta S_4^\circ - \Delta G^\circ_{vap}$